

gRPC MACH Architecture

TCO Networking Costs Calculator

Point-to-Point Integration Pattern · Protocol Buffer IDL · AWS Egress Pricing

✔ File 'complexUnaryRPCPattern.proto' uploaded successfully!



Phase 4 — TCO Networking Costs Report

Full cost breakdown including architecture tactics, unit economics, and stakeholder summary.

SERVICE UNDER ANALYSIS

order-retriever-system

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🔌 BUC1

⚡ 950 base RPS

👥 450 End Users

📁 —

APPLIED ARCHITECTURE TACTICS

Client-side LB Server-side LB Retry mTLS OAuth 2.0 + JWT



Applied Architecture Tactics & Patterns



INFORMATIONAL — NO RPS IMPACT

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
Client-side Load Balancing	—	Balances the emission of gRPC packets across server instances.	None
Server-side Load Balancing	—	Balances the reception of gRPC packets across backend replicas.	None
mTLS (mutual TLS)	0 reconnects/hr	Both client and server present X.509 certificates. CA validates both. 30: Max Payload overhead: bytes per message — RFC 8446 Because of RFC 8446 Section-	None

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
		5.2 * TLSCiphertext outer header: 5 bytes: * + 1-byte opaque_type=23 (uint8 ContentType) * + 2-byte legacy_record_version (uint16 ProtocolVersion) * + 2-byte length (uint16 length). * TLSInnerPlaintext prefix (AAD): * + 1-byte Content type and no padding. * AEAD Pre-record nonce: RFC 8446 Section-5.3 * + 8-byte length (64 bits explicit portion added during encryption). * AEAD tag: RFC 5116 Section-5.1 * + 16-byte tag appended to encrypted_record for AEAD_AES_128_GCM ciphertext is exactly 16 octets longer than its corresponding plaintext . Handshake is connection-scoped — negligible RPS impact at this reconnect rate. AWS ACM certificates are free.	
OAuth 2.0 + JWT — bearer token	Unary interceptor TTL 3600s 1 client(s)	JWT Bearer header RFC 7515 Section-7.1 BASE64URL(UTF8(JWS Protected Header)) '.' * BASE64URL(JWS Payload) '.' * BASE64URL(JWS Signature) adds 650 bytes typical due to Payload overhead RFC 7519 §3 and RFC 9101: 650 bytes per message: Algorithm Standard claims RS256 Header 80 Payload (compact/std/ext) 200/400/600 Signature 344 Dots 2 Total 626/826/1026 → 650 typical . The token travels in every request's Authorization metadata header. Response messages carry no JWT overhead. Validation enforced by the configured gRPC interceptor.	None

 RPS-IMPACTING — INCREASES NETWORKING COSTS

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
Retry	48% max retries	Retries gRPC packet emission under temporary outages. Increases networking costs as up to N additional packets may be sent per logical request.	+456 req/s



RPS Adjusted by Active Tactics

BASE RPS	EFFECTIVE RPS
950	1,406



Security Byte Overhead Applied to Message Sizes

TLS OVERHEAD / MESSAGE	JWT OVERHEAD / REQUEST	EFFECTIVE REQUEST SIZE
30 bytes (typical)	650 bytes (typical)	8705 bytes
30 / bytes typical	300 / 650 / 1200 bytes (min/typical/max)	
EFFECTIVE RESPONSE SIZE		
824 bytes		

MESSAGE TYPE	PROTO SIZE (BYTES)	EFFECTIVE SIZE (BYTES)	REQUESTS / MONTH	THROUGHPUT (GB/MONTH)	TRANSFER COST (USD/MONTH)
Request	8025	8705	4B	29,545.3557 GB	—
Response	794	824	4B	2,796.7114 GB	\$ 251.70

Total Cost of Ownership (TCO) Breakdown

Cost Component	DR / BC	Monthly Cost	% of TCO
AWS Egress (Response Transfer)	Networking	\$251.70	100.0%
Total Cost of Ownership (TCO)		\$251.70/mo	100.0%

* % of TCO is calculated against the sum of ALL cost rows shown above (must total 100%). DR = Disaster Recovery | BC = Business Continuity.



Unit Economics

TOTAL MONTHLY TCO (NETWORKING + CLOUD INFRA)

\$251.70

\$251.70 egress networking only



Per-Microservice Cost Metrics

TCO / MICROSERVICE / MONTH

\$251.70

This service · full TCO

COST / REQUEST

\$6.9066e-8

3,644,352,000 req/mo

ANNUAL PROJECTION

\$3020.40

12× monthly TCO

 Per-End-User Cost Metrics (450 end users)

TCO / END USER / MONTH

\$0.5593

\$251.70 ÷ 450 users

TCO / END USER / DAY

\$0.018644

Monthly ÷ 30

BUSINESS USE CASE

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Pattern cost attribution

 Revenue, ARPU & ROI Analysis

Revenue model: The value entered in Phase 1 is the *expected monthly revenue per end user* for this service. Total monthly revenue = that value × number of end users. **ARPU** = revenue per user per month. **ROI** = (total revenue – TCO) ÷ TCO. **Break-even users** = minimum end users needed to cover full TCO at current ARPU. Net Margin per user = ARPU – TCO per user. Present to Finance and Leadership to frame architecture cost as a revenue-generating investment.

TOTAL MONTHLY REVENUE

\$6750

\$15.00/user × 450 users

ARPU / MONTH

\$15

Revenue per end user / month

NET MARGIN / USER / MO

\$14.4407

ARPU – TCO/user

MONTHLY ROI

2581.8%

(Revenue – TCO) ÷ TCO

ANNUAL ROI

2581.8%

12× monthly projection

BREAK-EVEN USERS

17

users to cover full TCO

REVENUE / \$1 INFRA

\$26.82

Revenue efficiency

ANNUAL REVENUE

\$81.000

12× monthly revenue

 1 — IP Address Planning (CIDR Sizing)

The more redundancy you build into AWS (Multi-AZ, replicas, NAT Gateways, ENIs), the more private IP addresses are consumed. Plan your VPC CIDR block generously (e.g. /16 = 65,536 IPs) and **leave enough address space within each subnet CIDR for auto-scaling headroom**. A /24 subnet provides only 251 usable IPs — exhausted quickly in a multi-tier, auto-scaled architecture.

 2 — Multi-Tier Architecture in AWS Terms

Separating MACH components (frontend, API gateway, gRPC services, databases) into **different subnets** makes this a **multi-tier architecture** in AWS terms. Each tier needs dedicated subnets with its own Security Groups and NACLs.

 3 — Intra-VPC Connectivity

Use **VPC Endpoints** (PrivateLink) to access AWS services without internet routing; **Transit Gateway** for cross-VPC connectivity; **VPC Flow Logs** for visibility. Each component in a separate subnet requires explicit route table entries.

* Outbound (response) data transfer cost uses AWS standard internet egress tiered pricing for US East (N. Virginia): first 10 TB at \$0.09/GB, next 40 TB at \$0.085/GB, next 100 TB at \$0.07/GB, beyond 150 TB at \$0.05/GB. Inbound (request) traffic is free. Effective sizes include TLS (RFC 8446) and JWT (RFC 7519) overhead where applicable. [AWS Pricing Calculator](#) ↗



Phase 5 — Portfolio Summary

Aggregated TCO for all modelled microservices. Data persisted in browser local storage.

 Each service you calculate and add to the portfolio appears here with its description, Business Use Case, and total monthly TCO. Use **Add to Portfolio** in Phase 4. Data is stored in your browser's **local storage** — no server required, survives page reload.

MACH Architecture — TCO Portfolio Summary

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Service	BUC	Purpose	Monthly TCO	Annual TCO
order-management-system	BUC1	O M S	\$3917.74/mo	\$47012.88/yr
order-retriever-system	BUC1		\$251.70/mo	\$3020.40/yr

Service	BUC	Purpose	Monthly TCO	Annual TCO
TOTAL (2 services)			\$4169.44/mo	\$50033.28/yr

Architectural TCO estimate. Validate with Engineering, FinOps, Finance and Procurement before acting on these figures.